Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	**Semester**	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	**Batch**	2023–2027
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Experiment 6

Decision Tree and Random Forest: A Comparative Classification Study

Objective

- To implement a Decision Tree classifier.
- To extend it into a Random Forest ensemble model.
- To analyze hyperparameter impact using 5-Fold Cross-Validation.
- To compare single-tree and ensemble models.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Classes: Malignant (M), Benign (B)

Dataset Link: UCI Repository

Exploratory Data Analysis

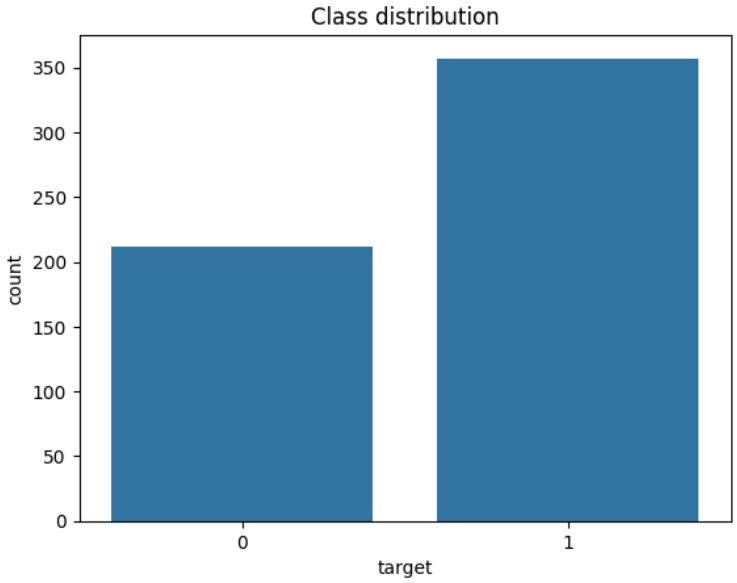

Figure 1: Class Distribution

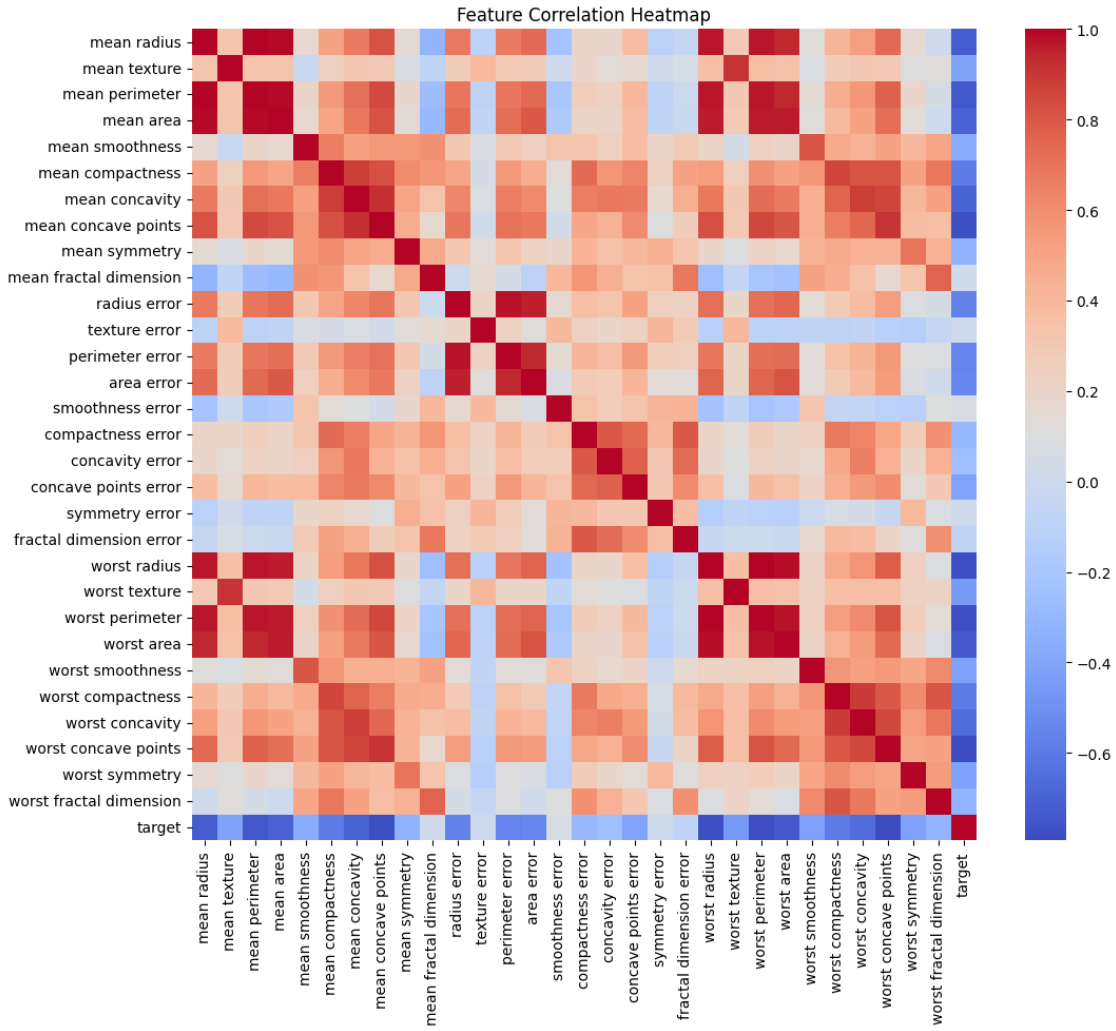

Figure 2: Feature Correlation Heatmap

Confusion Matrix and ROC

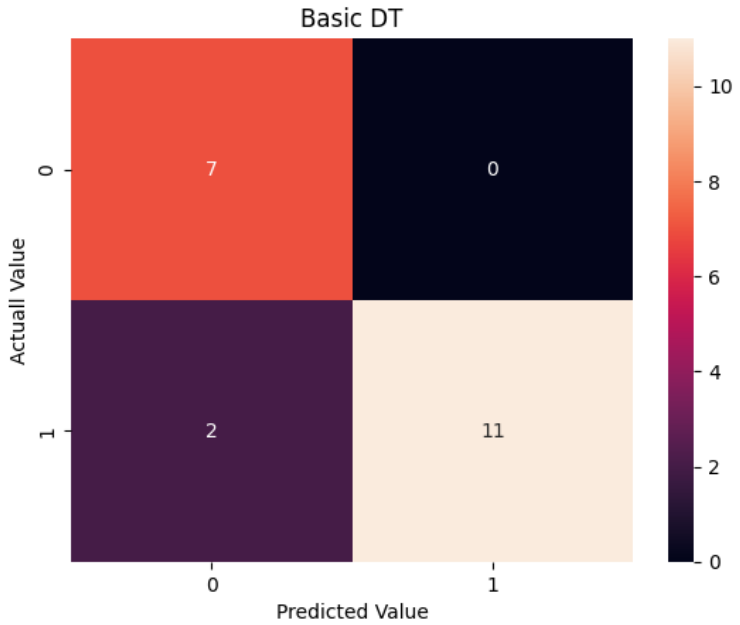

Figure 3: Decision Tree Confusion Matrix

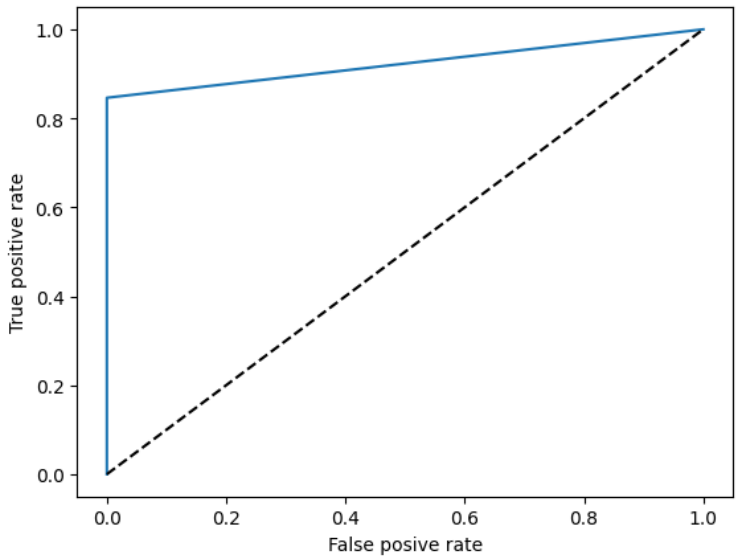

Figure 4: Decision Tree ROC Curve

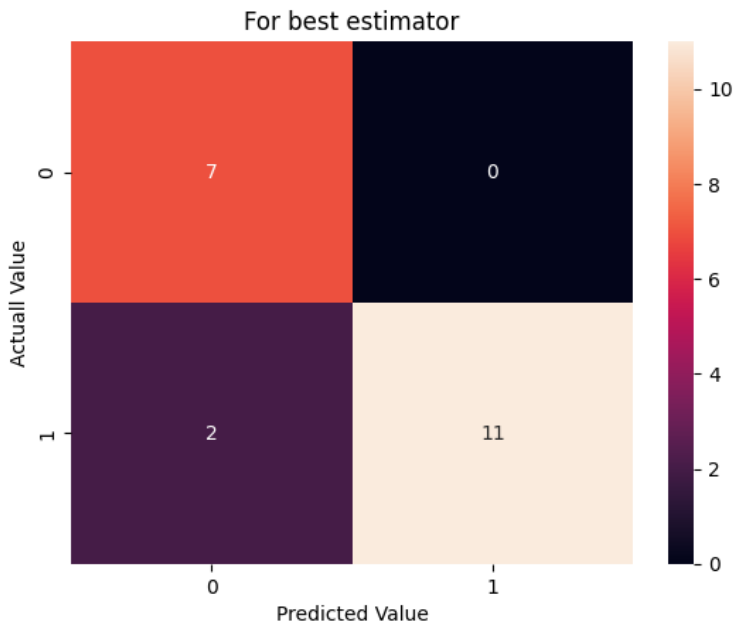

Figure 5: Random Forest Confusion Matrix

Hyperparameter Tuning Results

Decision Tree (5-Fold CV)

Table 1: Decision Tree Cross-Validation Results

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1
Gini	5	91.91	0.9606

Random Forest (5-Fold CV)

Table 2: Random Forest Cross-Validation Results

Estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1
200	5	sqrt	96.66	0.9755

5-Fold Cross-Validation Accuracy Comparison

Table 3: Model-wise CV Accuracy

Model	Fold1	Fold2	Fold3	Fold4	Fold5	Average
Decision Tree	89.47	91.23	92.98	94.74	91.15	91.91
Random Forest	94.74	93.86	98.25	98.25	98.23	96.66

Evaluation Metrics

Table 4: Test Set Performance

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	0.90	1.00	0.8462	0.9167
Random Forest	0.95	1.00	0.9231	0.96

Observations and Conclusion

- Decision Trees tend to overfit with increasing depth.
- Random Forest reduces variance using ensemble learning.
- Hyperparameter tuning significantly improves performance.

Conclusion: Random Forest achieved superior accuracy and generalization compared to Decision Tree, proving the effectiveness of ensemble methods.

References

- Scikit-learn Documentation
- UCI Machine Learning Repository